Impacts of LTPBR on Lower Elkhorn Creek

Kate McCarthy, Kyle Robinette, Chloe Rogozinski, Lynette Schwanger, Taia Storrs

Background

The Partnership: Collaborative research between Trout Unlimited, Walton Family Foundation, and the Colorado Water Conservation Board.

The Problem: Streams across the west have seen a decline in beaver populations, leading to a loss in surface and groundwater retention, and sediment capture.

Background

Restoration Strategy:

- "Low-Tech" Process-Based Restoration (LTPBR)
- Beaver mimicry structures (BMS)
- Post-assisted log structures (PALS)

Study Scope: 6 Colorado Headwater Sites

- Poudre River**: Lower Elkhorn**

Questions and Objectives

How do LTPBR treatments impact surface and ground water levels?

Objective:
Measure Surface and groundwater levels before and after LTPBR installation.

- Baseline surface and groundwater conditions
- Water levels before and after installation
- Changes in floodplain water storage

Questions and Objectives

How does the installation of LTPBR structures impact streamflow discharge?

Objective:
Measure streamflow discharge upstream and downstream before and after LTPBR structure installation.

- Reduce peak flows
- Increase drought and wildfire resiliency
- Attract beavers

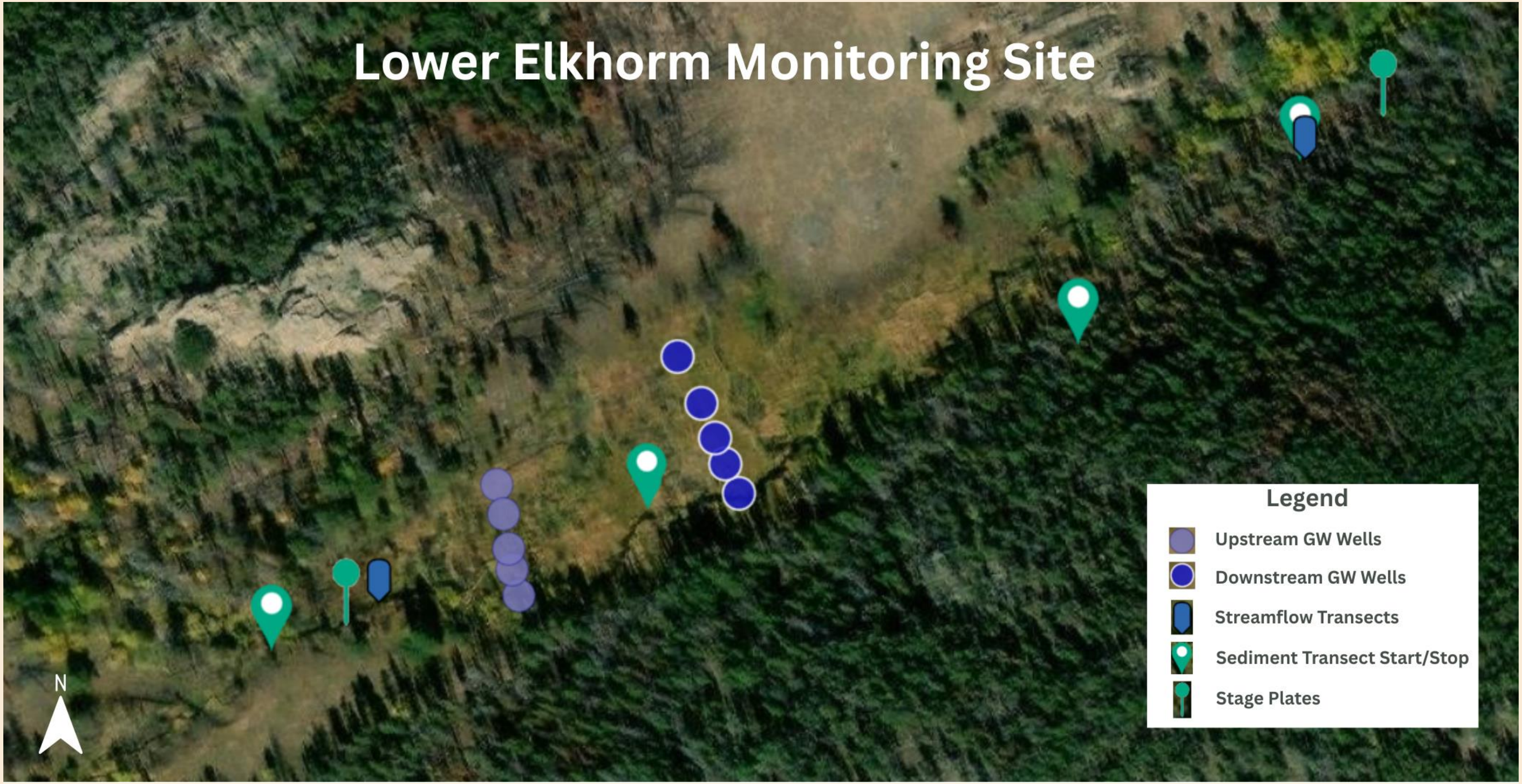
Questions and Objectives

How does the installation of LTPBR structures impact river sediment depth and deposition?

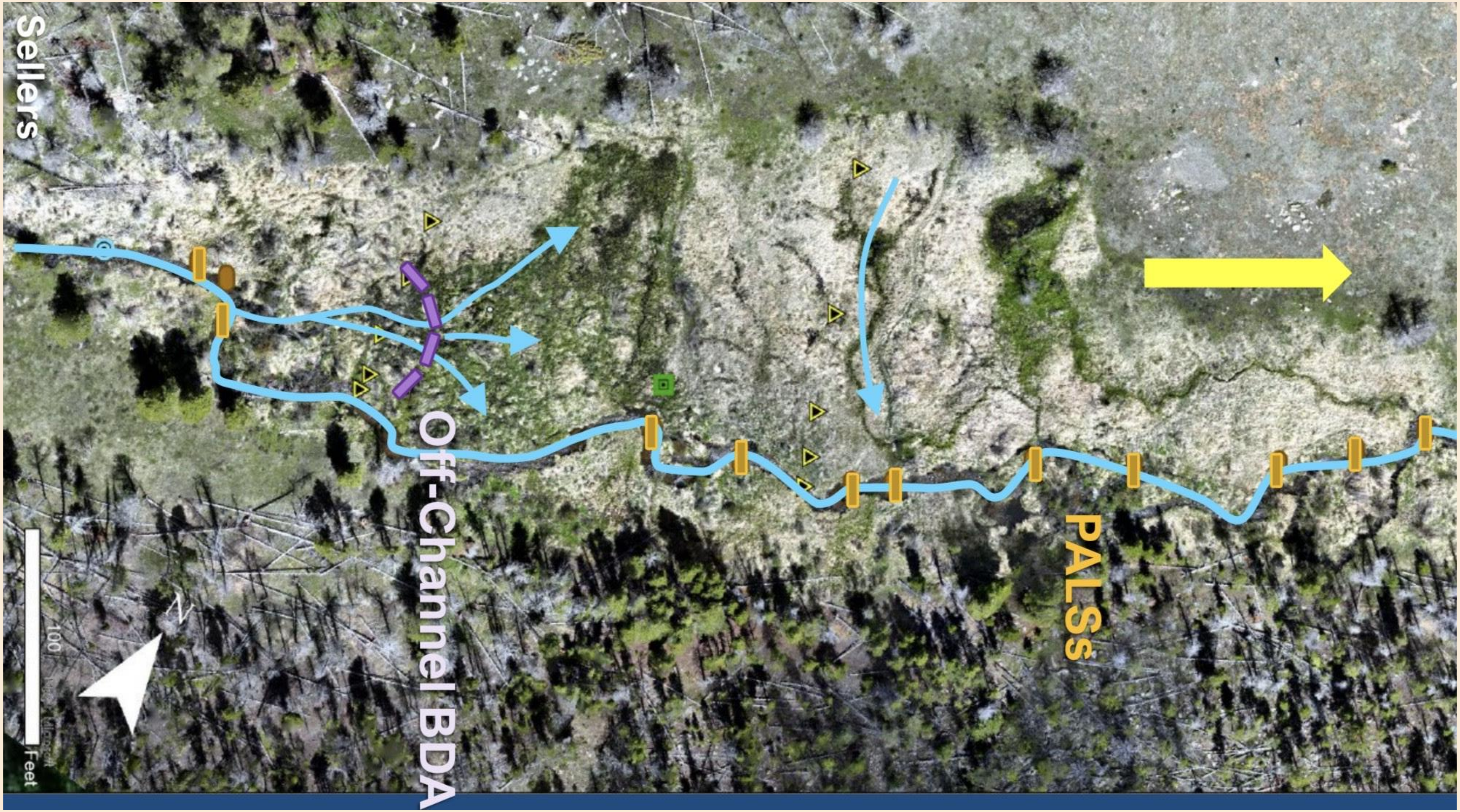
Objective:
Measure stream depth every 10 meters along the reach from the upstream sensor to the downstream sensors.

- Compare pre- and post-restoration sediment depths
- Trap sediment → Improve water quality

Site Location – Lower Elkhorn Creek



Site Location – Lower Elkhorn Creek – PALSs + BDA



Site Pictures – Lower Elkhorn Creek

PALS

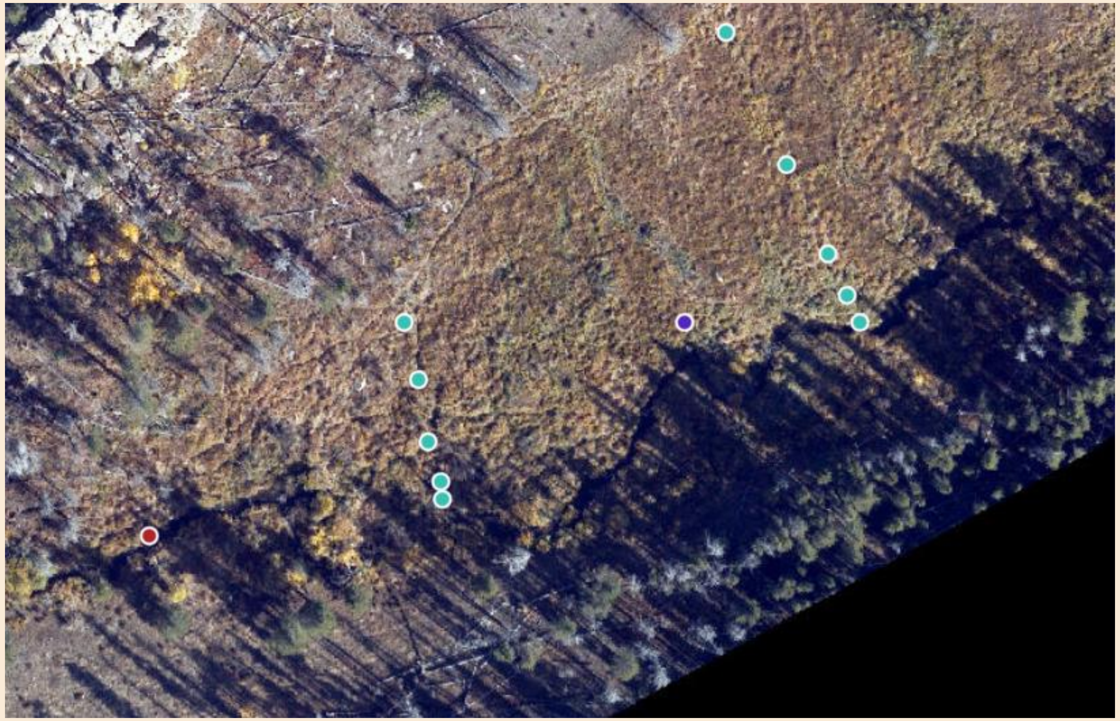
Stage Plate

Groundwater Wells

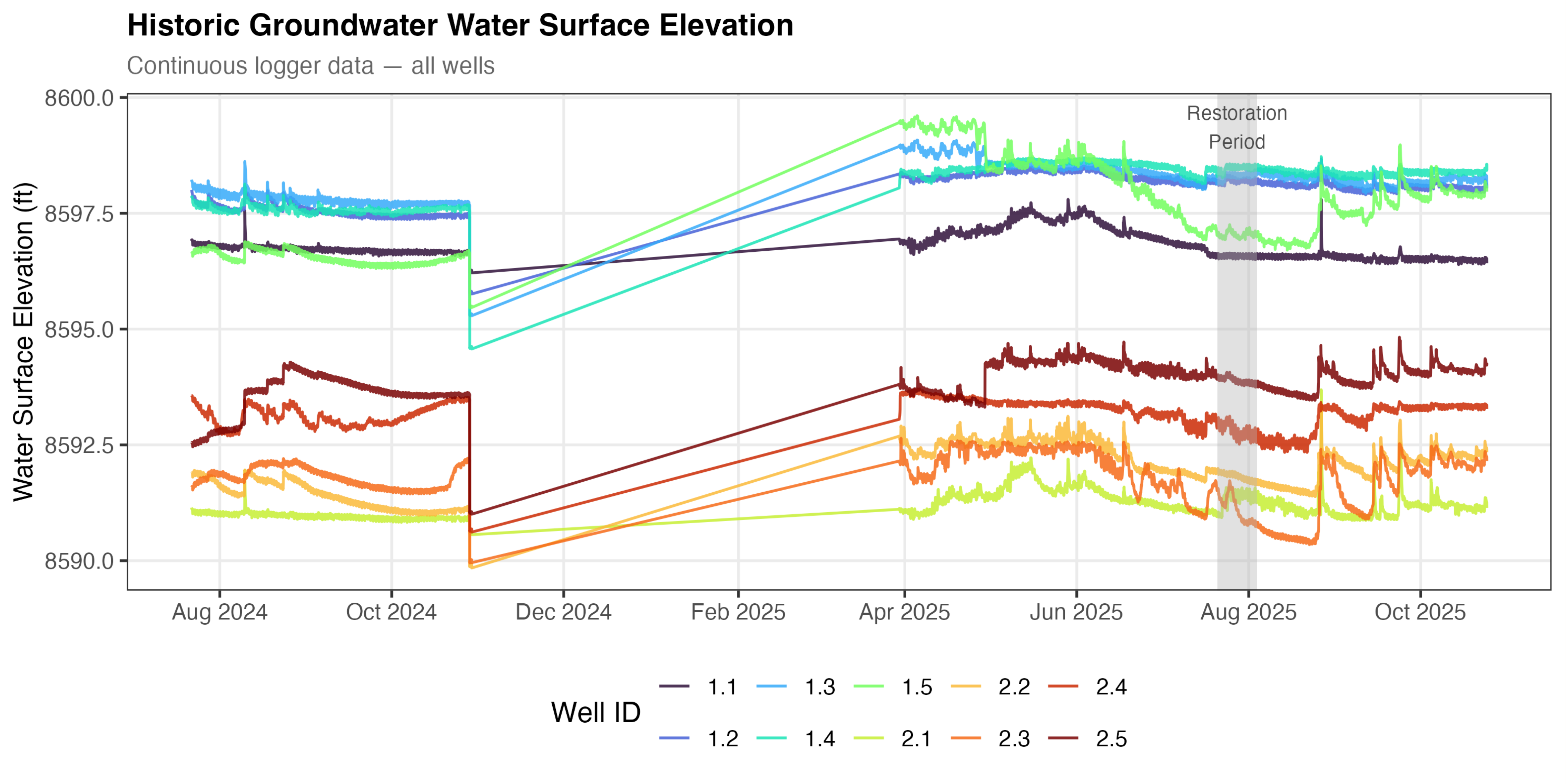
Streamflow Transect

Methods

Existing Project data → One upstream and one downstream site → Discharge (CFS), Stream Depth, Groundwater Depth, Sediment depth → Rstudio Data Analysis

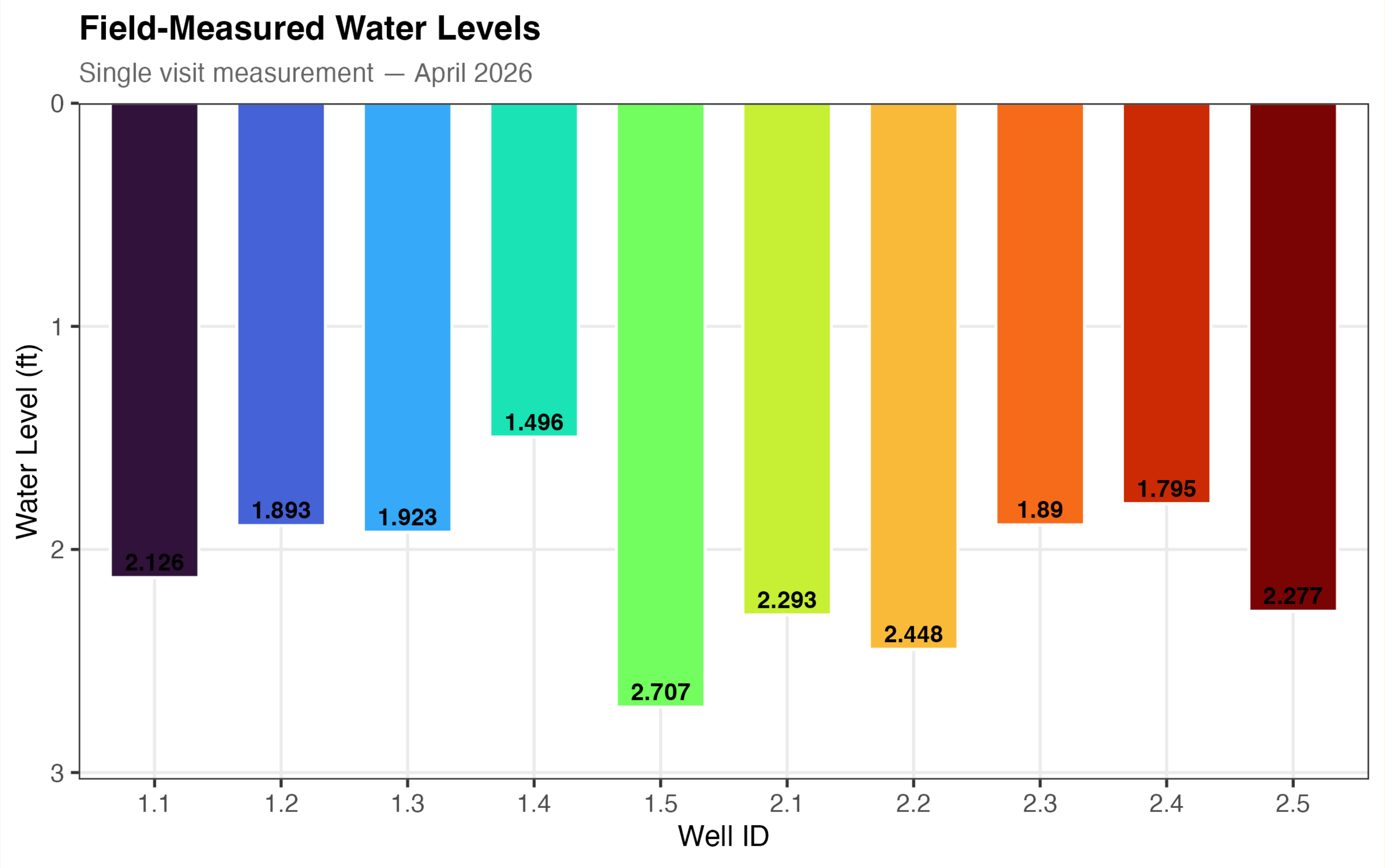


Objective 1 Results: Groundwater Well Monitoring

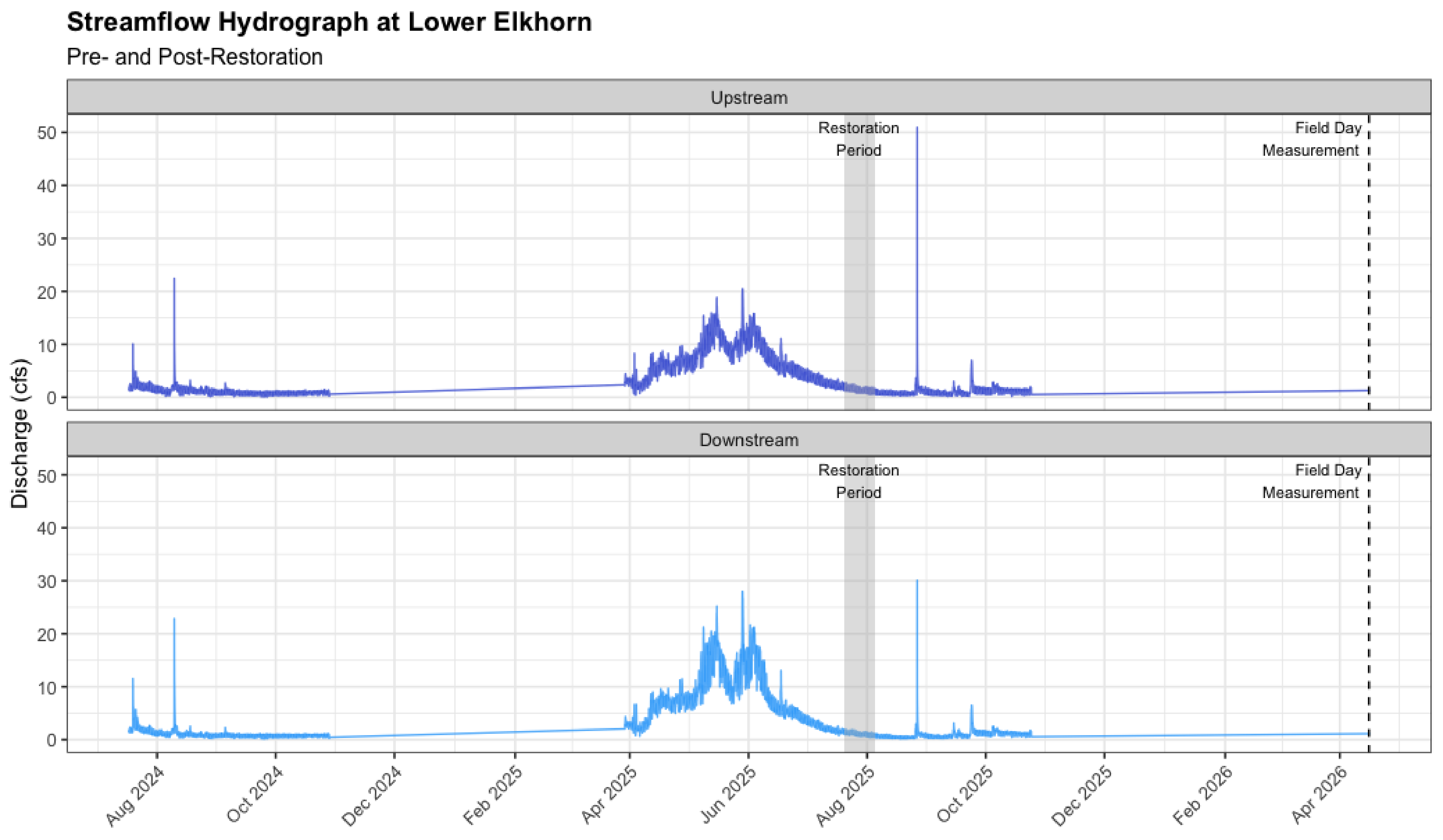


Data from Joel Sholtes

Objective 1 Results: Groundwater Well Monitoring



Objective 2 Results: Streamflow Discharge



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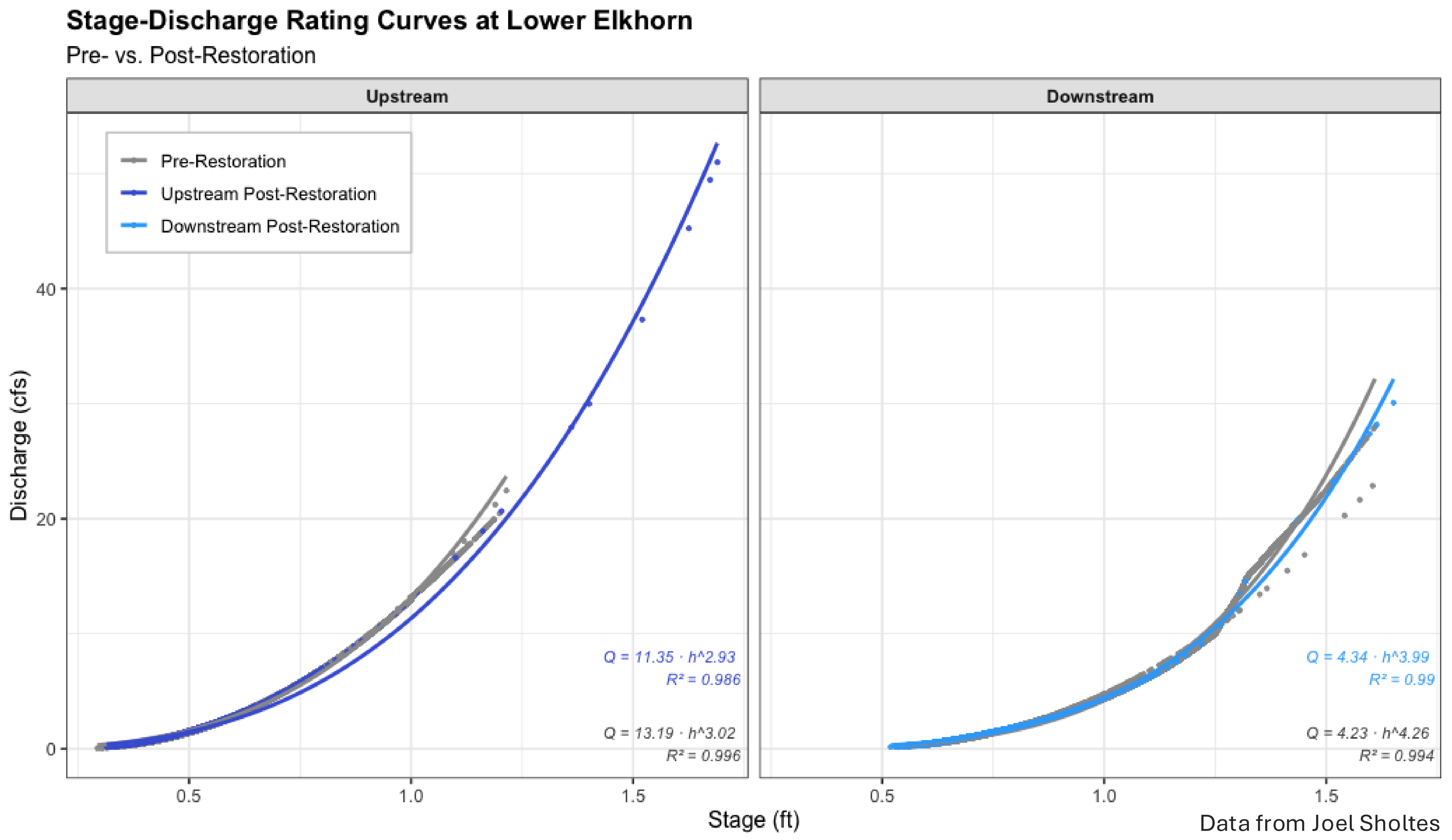
Objective 2 Results: Streamflow Discharge



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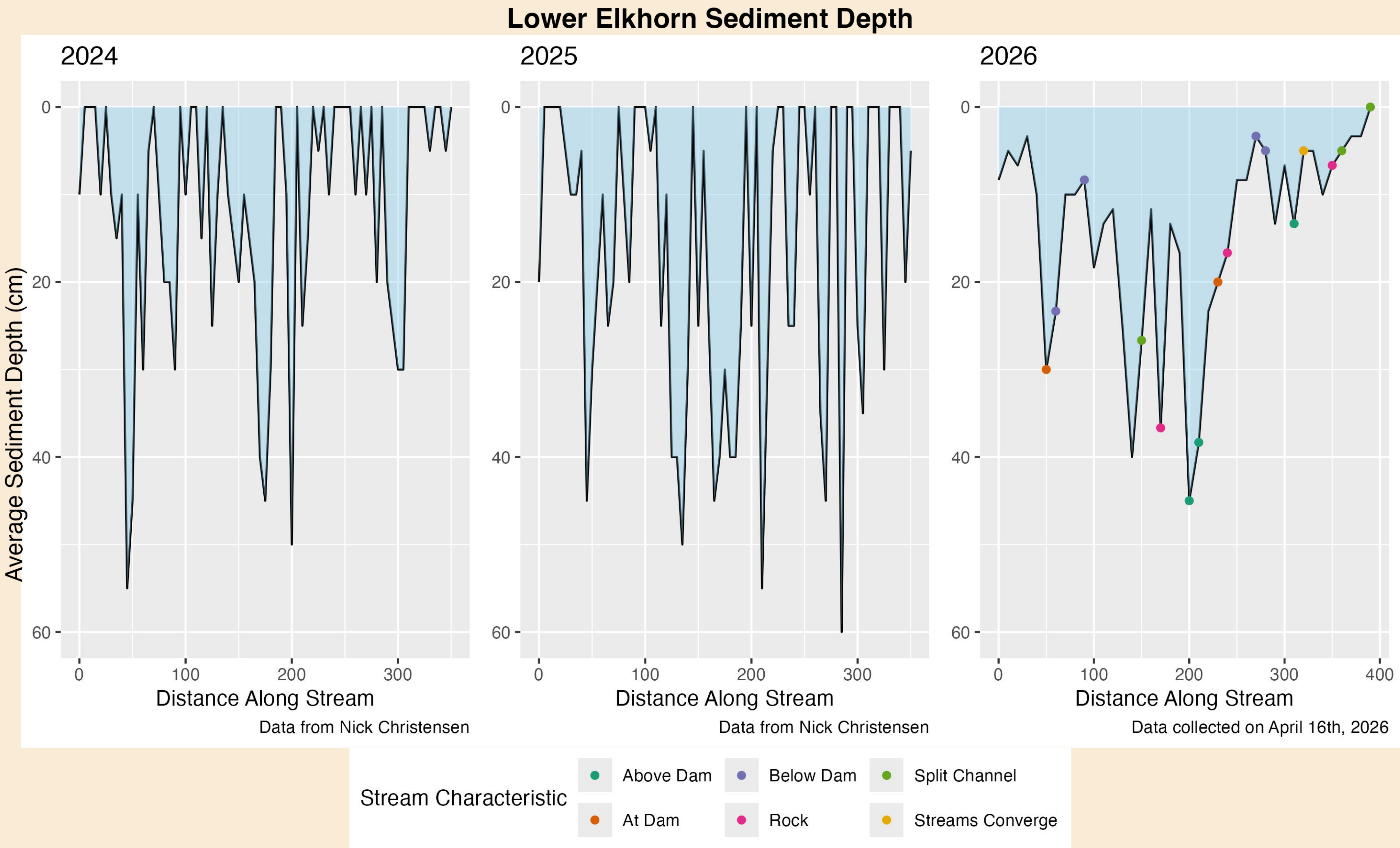
Objective 3 Results: Sediment Depth



Conclusion

- Early signals show PALS are influencing hydrology
- Trends suggest changes in groundwater, flow, and

sediment

- Dataset is still short-term
- More time + data needed to confirm effectiveness


QUESTIONS?
Thank you Kira, Nick, and Joel!